# Evaluation Study

Ablation study across six retrieval configurations

Model: `gemma4:e2b` | 20 ground-truth queries | Corpus: 176 dense vectors / 77 BM25 documents

| Metric | Dense Only | BM25 Only | Hybrid Static | Hybrid Adaptive | Hybrid + Reranker | Full Pipeline |
|---|---|---|---|---|---|---|
| **RETRIEVAL METRICS** | | | | | | |
| Embedding time ↓ | 12.67 s | **0.00 s** | 11.63 s | 12.32 s | 12.47 s | 12.31 s |
| Retrieval latency ↓ | 0.063 s | **0.004 s** | 0.048 s | 0.052 s | 0.169 s | 0.171 s |
| Relevance – Text ↑ | **0.7503** | 0.6031 | 0.7471 | 0.7458 | 0.7423 | 0.7431 |
| Relevance – Image ↑ | 0.4110 | **0.5773** | 0.4108 | 0.4043 | 0.4120 | 0.4042 |
| Page Coverage@k ↑ | 90.00% | 90.00% | 92.50% | **95.00%** | 90.00% | 90.00% |
| **GENERATION QUALITY** | | | | | | |
| ROUGE-1 ↑ | 0.5153 | 0.5074 | 0.5120 | 0.5138 | **0.5229** | 0.5124 |
| ROUGE-2 ↑ | 0.1976 | 0.1904 | 0.1998 | 0.1987 | **0.2123** | 0.2022 |
| ROUGE-L ↑ | 0.2878 | 0.2827 | 0.2938 | 0.2915 | **0.3056** | 0.2875 |
| BLEU ↑ | 0.1307 | 0.1257 | 0.1274 | 0.1329 | **0.1458** | 0.1352 |
| METEOR ↑ | 0.3353 | 0.3334 | 0.3304 | 0.3392 | 0.3466 | **0.3489** |
| BERTScore F1 ↑ | 0.8763 | 0.8748 | 0.8777 | 0.8770 | **0.8793** | 0.8772 |
| FCD ↓ | 17.87 | 18.56 | 18.29 | **16.96** | 19.09 | 18.89 |
| Faithfulness ↑ | 87.86% | 89.83% | **95.83%** | 92.20% | 93.57% | 92.95% |
| GT Coverage ↑ | 89.77% | 90.10% | 91.90% | 92.02% | **92.18%** | 92.07% |
| **PERFORMANCE** | | | | | | |
| End-to-end latency ↓ | 16.43 s | **8.37 s** | 16.05 s | 17.35 s | 17.32 s | 16.74 s |
| Throughput ↑ | 0.061 q/s | **0.120 q/s** | 0.062 q/s | 0.058 q/s | 0.058 q/s | 0.060 q/s |
| RAM usage ↓ | 0.125 GB | 0.930 GB | 0.125 GB | **0.111 GB** | 0.126 GB | 0.231 GB |
| GPU utilisation ↓ | 27.48% | 41.12% | 27.09% | **26.49%** | 26.97% | 27.32% |

### Configurations

**Dense Only** — semantic vector search only.
**BM25 Only** — lexical BM25 search only.
**Hybrid Static** — BM25 + semantic with fixed 0.5 / 0.5 weights.
**Hybrid Adaptive** — query-aware weight blending.
**Hybrid + Reranker** — static hybrid with cross-encoder reranking.
**Full Pipeline** — adaptive hybrid + reranker + distance filtering.

### Metric Notes

↓ lower is better | ↑ higher is better.

**FCD** — Faithfulness Cosine Distance; divergence between response and prompt embeddings.

**GT Coverage** — fraction of ground-truth answer tokens present in retrieved context.

Retrieval metrics are model-independent and computed once per configuration before any LLM call.

**Green cells** indicate the best (optimal) value for that metric.